

Conditional Probability:

Consider an experiment of rolling a 6-sided die.

Define two events:

- $A = \text{"roll is even"} = \{2, 4, 6\}$
- $B = \text{"roll is odd"} = \{1, 3, 5\}$

If we are told that event A occurred, can we say anything about B ?

Can event B occur?

→ No! A and B are disjoint, so B cannot occur.

What if we have $C = \text{"roll is less than 4"} = \{1, 2, 3\}$. If we know A occurred, can we say anything about C ?

→ C may or may not occur!

Goal: We would like to study how the probability of an event, e.g. C , changes when we are told that another event, e.g. A , occurred.

→ In other words, how likely is C conditional on A 's occurrence?

Def. (Conditional Probability): If A, B are events and $IP(B) > 0$,

then the probability of A given B (or conditional on B) is:

$$IP(A|B) = \frac{IP(A \cap B)}{IP(B)}.$$

Observation #1: If A and B are disjoint, then $P(A|B) = 0$. (Why?)

Observation #2: If $B = A$, then $P(A|B) = P(A|A) = 1$. (Intuitively, why?)

Observation #3: $P(A^c|B) = \frac{P(A^c \cap B)}{P(B)}$, and since $P(B) = P(B \cap A) + P(B \cap A^c)$,

by the law of total probability, then:

$$P(A^c \cap B) = P(B) - P(A \cap B)$$

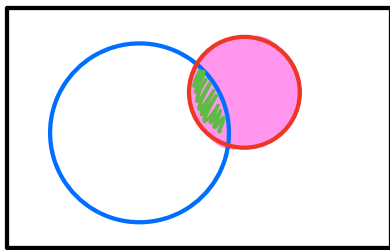
So:

$$P(A^c|B) = \frac{P(B) - P(A \cap B)}{P(B)} = 1 - \frac{P(A \cap B)}{P(B)}$$

Hence:

$$\underline{P(A^c|B) = 1 - P(A|B)}.$$

An illustration of conditional probability:



$$S = \mathcal{U}$$

$$A = \mathcal{U}$$

$$B = \mathcal{U}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{Area in both } A \text{ and } B}{\text{Area in } B}$$

Example: Last time, we saw that the prob. of a 5-card hand having exactly 2 aces was 4%.

Now, we want to compute $P(\underbrace{\text{Exactly 2 Aces}}_{\text{Event A}} \mid \underbrace{\text{At Least 1 Ace}}_{\text{Event B}})$

We know:

- $P(A) = 4\%$
- Total # of hands $\approx 2.6 \text{ M}$

To compute this, we need:

- $P(A \cap B) = P(\text{Exactly 2 aces AND At least 1 ace})$
→ Since $A \subset B$ (all hands with exactly 2 aces have at least 1 ace)
then $A \cap B = A$, so $P(A \cap B) = P(A) = \underline{4\%}$

$$\begin{aligned} \bullet P(B) &= 1 - P(B^c) = 1 - P(\underbrace{\text{No Aces}}_{\text{no aces}}) \\ &= 1 - 66\% = \underline{34\%} \\ &= \frac{\# \text{ hands w/ no aces}}{\text{total \# of hands}} = \frac{\binom{48}{5}}{2.6 \text{ M}} \approx 66\% \end{aligned}$$

$$\text{So: } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{4\%}{34\%} \approx \boxed{12\%} \leftarrow \text{Larger than } 4\%!$$

_____ ○ _____

Notice that we can write: $P(A \cap B) = P(A|B) \cdot P(B)$.

But by switching the order, if $P(A) > 0$, we also have:

$$P(A \cap B) = P(\text{B|A}) \cdot P(A).$$

Thm. (Bayes' Rule): If A, B are two events, then

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}.$$

Equivalently, we can write: $P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)}$

Proof: By law of total probability, $P(A) = P(A \cap B) + P(A \cap B^c)$

Plug in $P(A \cap B) = P(A|B) \cdot P(B)$ and $P(A \cap B^c) = P(A|B^c) \cdot P(B^c)$

to conclude. $\square$

We can also use the more general law of total probability to express $P(A)$ in the denominator of Bayes' Rule.

Thm. (General Bayes' Rule): If $B_1, B_2, \dots, B_n$ are mutually exclusive and collectively exhaustive events, then:

$$P(B_i|A) = \frac{P(A|B_i) P(B_i)}{\sum_{k=1}^n P(A|B_k) P(B_k)}$$

Example: Disease Testing.

Consider a disease that currently affects 5% of a population.

→ So, if we randomly chose an individual in this population,

the probability of the selected individual having the

disease is 5%. I.e. $IP(\text{Disease}+) = 5\%$. Hence, $IP(\text{Disease}-) = 95\%$.

We have a test, whose (random) outcome depends on whether the tested individual actually has the disease.

→ If Disease +, then the test has prob. 99% of being positive.

I.e. $IP(\text{Test}+ | \text{Disease}+) = 99\%$. Hence, $IP(\text{Test}- | \text{Disease}+) = 1\%$

→ If Disease -, then: $IP(\text{Test}- | \text{Disease}-) = 98\%$, and so

$IP(\text{Test}+ | \text{Disease}-) = 2\%$

Goal: Compute the likelihood that a randomly selected

individual who tested positive actually has the disease.

I.e., Compute $IP(\text{Disease}+ | \text{Test}+)$.

Calculations:

Before we start, make a guess!

- Equal to 99% ($= P(\text{Test}+ | \text{Disease}+)$)?

Start by using Bayes' Rule:

$$P(\text{Disease}+ | \text{Test}+) = \frac{P(\text{Test}+ | \text{Disease}+) \cdot P(\text{Disease}+)}{P(\text{Test}+)}$$

- $P(\text{Test}+ | \text{Disease}+) = 99\%$, given
- $P(\text{Disease}+) = 5\%$, given
- $P(\text{Test}+) = P(\text{Test}+, \text{Disease}+) + P(\text{Test}+, \text{Disease}-)$
 $= P(\text{Test}+ | \text{Disease}+) P(\text{Disease}+)$
 $+ P(\text{Test}+ | \text{Disease}-) P(\text{Disease}-)$
 $= 99\% \cdot 5\% + 2\% \cdot (1 - 5\%)$
 $= 6.85\%$

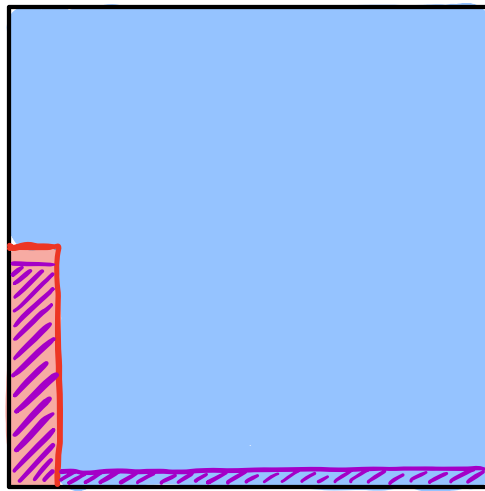
Putting the numbers together:

$$P(\text{Disease}+ | \text{Test}+) = \frac{99\% \cdot 5\%}{6.85\%} \approx \underline{\underline{72.3\%}}$$

→ Seems low... Lower than expected?

Illustration:

Imagine a population of 10000 people:



← 9500 don't have disease

• $9500 \cdot 2\% = 190$ of these test pos

• 9310 test negative

↑
500 have disease

{

- $500 \cdot 99\% = 495$ of these test positive
- $500 \cdot (1 - 99\%) = 5$ test negative

So there are $495 + 190 = 685$ total people testing positive, but among them only 495 truly have the disease.

→ Even though the test is "good", i.e. $P(\text{Test} + | \text{Disease} +)$ and $P(\text{Test} - | \text{Disease} -)$ are close to 1, the disease is very rare (5%). So, the # of individuals without the disease testing positive can be similar in size to the # of individuals with the disease that also test positive. If these are exactly equal, then $P(\text{Disease} + | \text{Test} +)$ would equal $1/2$!

Let's do the calculation again, but without specifying a number for $P(\text{Disease} +)$. Let this be equal to the variable p .

$$P(\text{Disease} + | \text{Test} +) = \frac{99\% \cdot p}{99\% \cdot p + 2\% \cdot (1-p)}$$

- p is close to 1? $\leadsto$ Prob. is close to 1
 - p is close to 0? $\leadsto$ Prob. is close to 0
-

Independence of Events:

The premise of conditional probability is that knowing some event A occurred can affect the likelihood that another event B also occurred.

But it may very well be the case that knowledge of A has no effect on the likelihood of B .

Def. (Independent Events): We say that events A and B are independent if: $P(A \cap B) = P(A) \cdot P(B)$.

If A and B are not independent, they are said to be dependent.

Observation #1: If $P(B) > 0$, A and B being independent implies that $\underline{P(A|B)} = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = \underline{P(A)}$.

I.e. A is just as likely regardless of whether we know B occurred.

Observation #2: If A and B are disjoint, and both have non-zero probability, then they are dependent events.

Why? $P(A \cap B) = P(\emptyset)$, by A, B disjoint
 $= 0$, by properties of prob.

But $P(A) \cdot P(B) \neq 0$ by assumption.

Example: Roll 4-sided die

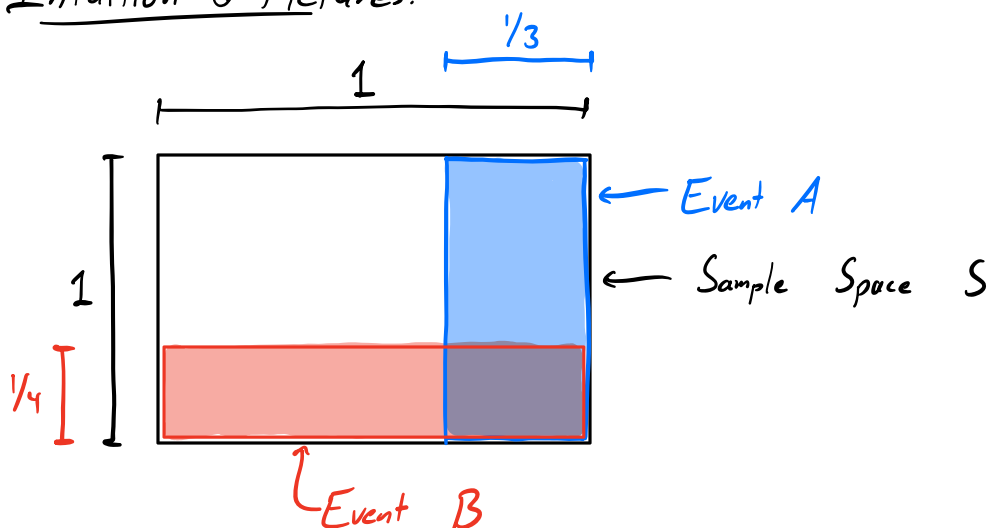
$S = \{1, 2, 3, 4\}$, with equally likely outcomes.

Let: $A = \{1, 2\}$ and $B = \{1, 3\}$

Notice: $P(A) = 1/2$, $P(B) = 1/2$, and $P(A \cap B) = P(\{1\}) = 1/4$

$\leadsto$ A and B are independent events!

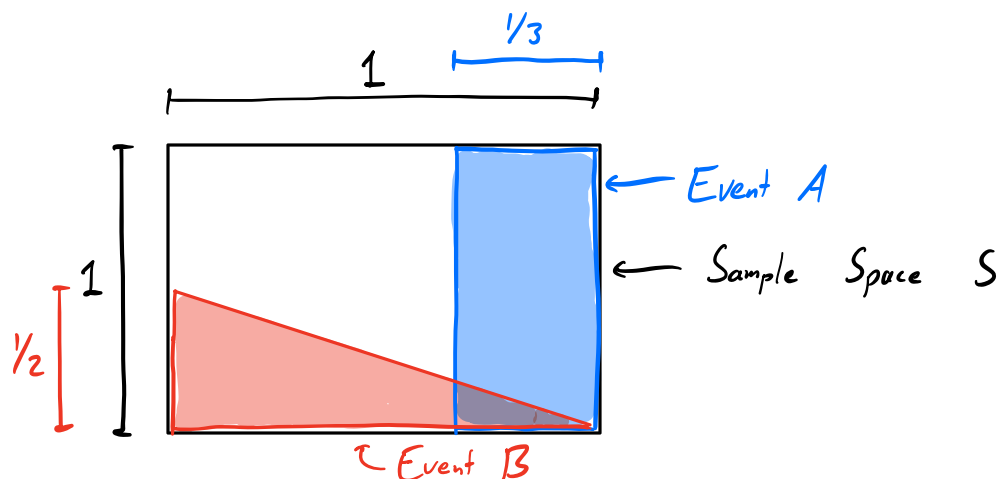
Intuition & Pictures:



Note: In this example, probability of an event = area

$$P(A \cap B) = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12} \quad \leftarrow \text{Equal, so } A, B \text{ are independent events!}$$

$$\left. \begin{array}{l} P(A) = 1/3 \\ P(B) = 1/4 \end{array} \right\} \Rightarrow P(A) \cdot P(B) = \frac{1}{12}$$



Still, $P(A) = 1/3$ and $P(B) = 1/4$

But: $P(A \cap B) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{6} = \underline{\underline{1/36}}$

$P(A) \cdot P(B) = \underline{\underline{1/12}}$ $\leftarrow$ Not Equal! A, B are not independent events.

We can also generalize the notion of independent events to settings where we have more than 2 events.

Def. (Pairwise Indep.): Events $A_1, A_2, \dots, A_n$ are pairwise independent if for any $1 \leq i < j \leq n$, events A_i and A_j are independent.

Def. (Mutual Independence): Events $A_1, A_2, \dots, A_n$ are mutually independent if for any $k = 2, 3, \dots, n$, and every subset of size k , denoted

$A_{i_1}, A_{i_2}, \dots, A_{i_k}$ (where $i_1, i_2, \dots, i_k$ are in $\{1, 2, \dots, n\}$), we have

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \cdot P(A_{i_2}) \cdot \dots \cdot P(A_{i_k})$$

Note: Mutual independence is stronger than (i.e. implies)

pairwise independence. But pairwise indep. $\nRightarrow$ mutual indep.

Example: Pairwise but not Mutually Indep. Events

$S = \{1, 2, 3, 4\}$, with equally likely outcomes.

Let: $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{2, 3\}$

For Pairwise Indep., we need:

$$\bullet P(A \cap B) = P(A) \cdot P(B) \quad \checkmark \text{ (shown above)}$$

$$\bullet \underbrace{P(A \cap C)}_{\substack{\{2\} \\ 1/4}} = \underbrace{P(A)}_{1/2} \cdot \underbrace{P(C)}_{1/2}$$

$$\leadsto 1/4 = 1/2 \cdot 1/2 \quad \checkmark$$

$$\bullet \underbrace{P(B \cap C)}_{\substack{\{3\} \\ 1/4}} = \underbrace{P(B)}_{1/2} \cdot \underbrace{P(C)}_{1/2}$$

$$\leadsto 1/4 = 1/2 \cdot 1/2 \quad \checkmark$$

$\Rightarrow A, B, C$ are pairwise independent

For Mutual Independence, we also need

$$\underbrace{P(A \cap B \cap C)}_{\substack{\emptyset \\ 0}} = \underbrace{P(A)}_{1/2} \underbrace{P(B)}_{1/2} \underbrace{P(C)}_{1/2}$$

$$\leadsto 0 \neq 1/2 \cdot 1/2 \cdot 1/2$$

$\Rightarrow A, B, C$ are not mutually independent